

FIT3163 Project Proposal Report

Social Media Regional Housing Intelligence

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Authorship statement

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Fauzanda wrote the initial drafts for Section 1 (Introduction) and Section 2 (Background Material/Literature Review). He finalised the text for Section 2 as well. He prepared the RTM (Appendix C).

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Vincent wrote the initial drafts and finalised the text for Section 3.1 (Project Scope), Section 3.2 (Project Requirements), Section 3.5 (Project Schedule), and Section 3.6 (Project Resources). He prepared Table 1 (In-Scope Items), Table 2 (Out-of-Scope Items), Table 3 (Product Acceptance Criteria), the WBS (Appendix A), the Gantt chart (Appendix B), and User Survey (Appendix F).

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Priscilla wrote the initial draft for Section 4 (Methodology) and Section 5 (Conclusion), and finalised the text for Section 1, 4, and 5. She prepared Table 6 (List of Required Tools and Technologies), Table 7 (Example Keywords used for Topic Extraction), Table 8 (Sentiment Score Threshold), Figure 1 (System Architecture), and the Wireframe (Appendix E). She is also responsible for proofreading and style wrangling.

1 Introduction

Social Media Regional Housing Intelligence is a data science project developed by Fauzanda, Vincent, Danial, and Priscilla. It supports Melbourne homebuyers aged 30 to 45 by analysing public Reddit discussions and presenting suburb-level housing sentiment through an interactive dashboard.

1.1 Project Background

Metropolitan Melbourne is undergoing major urban and socio-economic change. According to the Australian Bureau of Statistics, Melbourne leads the nation in demographic expansion, with an annual population increase of approximately 105,000 residents (ABS, 2025). Economic forecasts also project a 7% rise in median residential property values across both detached houses and high-density units (KPMG, 2026).

For homebuyers, this creates a competitive and high-stakes market. Buying a home is not only a financial decision, but also a long-term lifestyle decision. However, traditional property data may not fully capture suburb-level differences or everyday living experiences. Public perception may shift due to affordability, safety, education and employment access, or lifestyle amenities, but these changes may not appear immediately in formal property data.

Digital communication offers an opportunity to understand local sentiment. Recent figures show that over 95% of Australians active online have used a digital communication network or social media platform within a typical six-month period (ACMA, 2023). Online discussions therefore provide public text data about suburb safety, amenities, affordability, and local living experiences.

1.2 Problem Statement

Although property websites and official datasets provide useful housing information, they mainly present structured indicators such as prices, listings, and demographic statistics. They may not clearly show how people currently perceive individual suburbs. As a result, homebuyers may need to manually search online discussions to understand local concerns and strengths, which is time-consuming and difficult to compare.

1.3 Project Aims and Objectives

This project aims to develop a web-based dashboard that analyses public Reddit posts and comments from r/melbourne. It focuses on 10 selected suburbs: Collingwood, Oakleigh, Nunawading, Eaglemont, Richmond, Carlton, Vermont, Dandenong, Officer, and Clayton.

The system will collect suburb-related Reddit discussions, clean and preprocess the text, classify discussions into four topic categories, apply sentiment analysis, and present the results visually. The topic categories are lifestyle amenities, education and employment, crime and safety, and affordability.

The expected outcome is a dashboard showing each suburb's sentiment score, sentiment breakdown, number of analysed posts, key concerns, and key strengths. It aims to complement traditional property data by providing a local, perception-based view of Melbourne suburbs.

1.4 Overview of Report Structure

Section 2 presents the Literature Review and identifies the research gap.

Section 3 outlines the Project Management Plan, including scope, schedule, risks, and team responsibilities.

Section 4 explains the Methodology, including data collection, sentiment analysis, system architecture, and dashboard design.

Section 5 concludes the report by summarising the project's expected contribution.

Sections 6 and 7 provide the Appendices and Bibliography.

2 Background Material / Literature Review

Introduction

Sentiment analysis refers to the automated process of extracting and summarising subjective opinions and emotional signals from large-scale textual data sources (Bordoloi & Biswas, 2023). One such type is location-based sentiment analysis, which employs natural language processing or machine learning algorithms to extract and classify the sentiment expressed within a body of text, anchoring those findings to the geographic source from which the text was produced (Almatrafi et al., 2015). This field has gained considerable momentum in recent years, fueled by the widespread availability of geo-tagged social media data and the growing sophistication of natural language processing tools. A 2026 review synthesising 20 studies on integrating large language models (LLMs) with geospatial social media analytics found that LLMs enhance analysis in disaster management, urban planning, and the study of human behaviour by enabling location inference from unstructured text, extending analysis beyond explicitly geotagged posts (Farhangi et al., 2026). Addressing the historical limitation that only a small fraction of social media posts carry precise geographic coordinates, researchers trained a BERT-based multilingual sentiment classification model on a global geotagged Twitter archive of 4.37 billion tweets, producing the Twitter Sentiment Geographical Index (TSGI) — an open-access database providing sentiment measurements across 164 countries at the county/city level on a daily basis (Chai et al., 2023).

Literature Review Body

Our project builds upon foundational studies that leverage unstructured text data to evaluate urban environments and real estate dynamics. A notable precedent is the Monash University project "Evaluating Neighbourhood Sentiment during Melbourne's Covid-19 Lockdowns" (Grodach et al., 2020), which analysed 2020 Twitter data across more than 300 Melbourne suburbs to examine localised public sentiment relative to specific neighbourhood features, architectural layouts, and local amenities. By identifying correlations between the built environment, community satisfaction, and emotional outlooks during pandemic restrictions, the researchers mapped these trends across space and time using an interactive digital dashboard and the Melbourne Digital City spatial modelling platform. However, a key limitation of this approach is that it only captures aggregate macro-sentiment and does not pinpoint the granular, specific drivers behind why a particular suburb exhibits a positive or negative emotional trajectory.

A second relevant study, "Reputation matters: residents' sentiment and housing price" (Choi et al., 2025), shifts focus toward the economic implications of resident perception, exploring resident sentiment as an explanatory variable in property valuation. Using dictionary-based textual analysis alongside numeric user ratings, the authors integrated sentiment proxies into traditional hedonic pricing models and advanced machine learning frameworks such as Random Forest and XGBoost. Findings indicated that combining text-driven sentiment indices with numeric ratings yields significantly higher explanatory power for housing prices than ratings alone, with SHAP values confirming that sentiment metrics remain statistically significant even when controlling for standard structural and spatial variables. Despite these contributions, this study primarily focuses on price forecasting and lacks an in-depth explanation of the distinct localised factors driving positive or negative sentiments within specific suburbs.

Conclusion

When synthesised, these two bodies of literature reveal a critical research gap: while existing studies successfully track general geographic sentiment trends over time or link aggregate sentiment to macroeconomic property valuations, neither approach deeply investigates the specific

underlying causes of sentiment within individual suburbs. Addressing this unmet need defines the core objective of our project.

The project employs a Python-based technical stack selected for its alignment with established practices in social media analytics. Reddit data is collected via PRAW (Python Reddit API Wrapper), which provides structured programmatic access to Reddit's API (Boe, 2014), while suburb-level demographic context is drawn from the ABS 2021 Census of Population and Housing. Sentiment classification is performed using VADER, a lexicon and rule-based tool purpose-built for short-form social media text that produces compound polarity scores without requiring labelled training data (Hutto & Gilbert, 2014). The frontend dashboard is built in React.js with Chart.js for interactive visualisation, and the backend is developed using Flask, with data persisted in PostgreSQL. Exploratory analysis is conducted in Jupyter Notebooks (Kluyver et al., 2016), with Git and GitHub managing version control throughout.

Regarding tool selection, VADER has been widely adopted in social media sentiment analysis due to its strong performance on short, informal text (Hutto & Gilbert, 2014). While transformer-based models such as BERT have demonstrated superior accuracy on formal text classification tasks (Devlin et al., 2019), comparative studies have found that VADER performs comparably on informal social media content such as Reddit posts, where the brevity and colloquial nature of text reduces the advantage of contextual embeddings (Elbagir & Yang, 2019). Furthermore, VADER's rule-based transparency and low computational overhead make it more appropriate for projects operating under resource and time constraints, with lexicon-based tools applied to Reddit data reporting consistent and interpretable results at the aggregate level (Medhat et al., 2014). On this basis, VADER represents a justified and practical choice over more computationally demanding alternatives for the scope and scale of this project.

The project adopts an Agile SDLC using a lightweight Scrum framework, selected over Waterfall due to the exploratory nature of the work, particularly the need to experiment with sentiment analysis techniques and dashboard usability before stable requirements could be fixed. Unlike Waterfall's sequential structure, Agile allows requirements to be continuously refined throughout development (Anurina, 2021), with Scrum's iterative approach enabling incremental feature delivery, ongoing user feedback, and early MVP release (Dhaduk, 2022). The team adapted standard Scrum to suit academic scheduling constraints by replacing daily stand-ups with milestone meetings, Trello-based task tracking, and ongoing online communication — an approach supported by research showing that lightweight, tool-mediated coordination helps reconcile Scrum with academic obligations (Damianova & Berrezueta-Guzman, 2026). Scrum is particularly well-suited to academic environments where project dimensions are unpredictable, as it reduces risk by focusing teams on smaller, more attainable tasks (Fernandes et al., 2021).

3 Project Management Plan

3.1 Project Scope

The Social Media Regional Housing Intelligence Dashboard is a web-based analytical tool that aggregates historical Reddit-derived sentiment data alongside ABS census demographics for 10 selected Melbourne suburbs. The intended user is a non-technical adult aged 30–45 researching suburbs for housing or lifestyle decisions. Scope boundaries have been defined and held consistent since the Pitch presentation.

In scope

Table 1: In-Scope Items		
Item	Priority	Rationale
NLP sentiment scoring across 10 categories per suburb	Critical	Core deliverable of the entire system
Overall suburb sentiment score with post/comment count	Critical	Enables users to assess data legitimacy
Multi-suburb side-by-side comparison view	Critical	Primary use case for the target demographic
Interactive web-based dashboard for non-technical users	High	Directly serves the 30–45 y.o. target user
Top strengths and concerns surfaced per suburb	Medium	Adds interpretability beyond raw scores
ABS census demographics alongside sentiment scores	Medium	Provides housing context for suburb comparisons
Scalable pipeline supporting additional suburbs	Low	Future-proofs the system without requiring redesign

Out of scope

Table 2: Out-of-Scope Items	
Item	Reason
Real-time or live Reddit monitoring	Technical complexity and API rate-limit constraints
Property price prediction or forecasting	Outside defined analytical task; risk of misleading users
Coverage beyond 10 selected Melbourne suburbs	Infeasible within one-semester timeline and resources
Multilingual content analysis	Significant NLP complexity beyond current project scope
Causal inference modelling	Requires external datasets and advanced statistical methods

Assumptions and constraints

The project assumes selected suburbs generate sufficient Reddit post volume for statistically meaningful indices and that Reddit API access remains stable within free-tier rate limits throughout the semester. ABS 2021 census data is assumed sufficiently current for suburb-level context. Key constraints include a 12-week semester timeline, a 4-person team, no dedicated budget, and a requirement that all outputs carry financial advice disclaimers under Australian consumer law.

3.2 Project Requirements

Functional requirements centre on giving users meaningful, transparent data. The system must display a sentiment score for each suburb alongside the volume of posts and comments underpinning it, so users can judge the reliability of each score. It must also surface the top strengths and concerns per suburb derived from topic analysis, and support side-by-side comparison of two suburbs to aid decision-making. Users can select from a predefined set of Melbourne suburbs, bounding the tool to its defined coverage area.

Non-functional requirements prioritise accessibility for the target demographic. A user survey conducted with prospective users in the 30–45 age group informed the interface requirements, with respondents indicating that clarity of information presentation and ease of navigation were the most important factors when using an unfamiliar data tool ([Appendix F](#)). The interface must present a clear visual hierarchy so users can identify key information at a glance, and must be intuitive enough for non-technical users to navigate without guidance.

ID	Acceptance Criteria
01	Post/comment count is visible alongside every category score for all 10 suburbs.
02	Sentiment scores are shown for all defined topic categories for each suburb.
03	Overall score is visible for all 10 suburbs and updates when underlying scores change.
04	Users can select 2 suburbs and view their scores side-by-side on a single page without page reload.
05	Selecting a topic filter updates the displayed data to show only the relevant category scores.
06	Suburbs are displayed in ranked order by overall sentiment score.
07	Sentiment scores for selected suburbs are visible on one page without navigation.
08	Searching a suburb name returns its overall score within 3 seconds.
09	Post and comment count is visible on the suburb result page alongside the sentiment score.
10	Target users can identify key suburb information within 10 seconds of landing on the page without guidance.

Table 3: Product Acceptance Criteria

The full Requirements Traceability Matrix is provided in [Appendix C](#).

3.3 Project Team Organization and Roles

Project Management Methodology

This project will use the Agile Scrum methodology. Agile was selected because the project involves data collection, sentiment analysis, and dashboard development, where requirements and technical challenges may evolve throughout the project. Agile enables the team to develop the system incrementally, obtain feedback regularly, and respond quickly to issues identified during testing.

The project will be completed over a 10-week period using short development iterations. Progress will be monitored continuously and features will be delivered incrementally as a Minimum Viable Product (MVP) before further refinement.

The following project management tools will be used:

- Trello – task allocation and progress tracking through a Kanban board.
- GitHub – version control and collaboration.
- Gantt Chart – monitoring project milestones and schedule.
- Meeting Minutes – documenting decisions, actions, and progress.

Team Organisation

Project responsibilities are distributed among team members according to the Work Breakdown Structure (WBS).

Team Member	Role	Key Responsibilities
Fauzanda	Backend Developer	Frontend Developer & Tester, Backend to Front end integration
Danial	Sentiment Analysis Developer	VADER implementation, sentiment scoring, model evaluation, Scrum master
Vincent	Data Engineer	Data collection, preprocessing, cleaning, validation
Priscilla	Frontend Developer & Tester	Frontend Developer & Tester, Backend to Front end integration

Table 4: Team Member Roles and Key Responsibilities

This allocation of responsibilities ensures accountability, balanced workload distribution, and efficient project execution.

3.4 Project Management Process

Risk Management

Risk management is a continuous activity throughout the project lifecycle rather than a one time planning exercise. Regular risk reviews will be conducted during team meetings to identify emerging risks, evaluate the effectiveness of mitigation strategies, and update the Risk Register

where necessary. This proactive approach helps minimise project disruptions and increases the likelihood of delivering the project within scope, budget, and schedule constraints.

The process consists of:

1. Risk Identification – Risks are identified during planning meetings and project reviews.
2. Risk Analysis – Risks are assessed based on likelihood and impact.
3. Risk Response Planning – Mitigation strategies are developed for significant risks.
4. Risk Monitoring – Risks are reviewed regularly and updated in the Risk Register.

The Risk Register will be maintained throughout the project and reviewed during team meetings.

Communication Plan

Effective communication is essential for project success. The following communication methods will be used:

Stakeholder	Method	Frequency	Purpose
Project Team	Team Meetings	5 meetings across project duration	Progress review and decision making
Project Team	Trello Board	Continuous	Task tracking and workload management
Project Team	Messaging Platform	As required	Day-to-day communication
Teaching Staff/Tutor	Consultation and Email	As required	Feedback and clarification
Stakeholders	Project Deliverables and Presentations	Major milestones	Progress reporting

Table 5: Communication Plan

Meeting minutes will be maintained to document decisions, action items, and responsibilities.

Monitoring and Control

Project progress will be monitored against the WBS and Gantt Chart milestones. Trello will be used to track task completion and identify delays. GitHub will provide version control and an audit trail of project changes.

Regular reviews will be conducted to ensure deliverables meet project requirements and quality standards. Testing activities, including functional, performance, and user acceptance testing, will be used to verify project outcomes.

3.5 Project Schedule

Activities and Tasks Overview

The project is structured across eight phases spanning twelve weeks, beginning 27 July 2026. The phases progress logically from initial setup through to final delivery, with data collection and frontend wireframing running in parallel during the early weeks to maximise efficiency.

The eight phases are: Planning & Setup, Data Collection, Frontend Development, Data Analysis & NLP, Backend Development, Frontend Integration & Visualisation, Testing & Refinement, and Finalisation. Early phases establish the data pipeline and UI skeleton, while mid-project work centres on the NLP sentiment analysis pipeline, the critical path of the project. Backend development and frontend integration follow once the analysis outputs are stable, with the final weeks dedicated to system testing, polish, and documentation.

A full Work Breakdown Structure detailing each phase and its constituent tasks is provided in [Appendix A](#).

Gantt Chart

The project timeline is visualised in a Gantt chart provided in [Appendix B](#). The chart maps all tasks across the 12-week schedule, reflects task dependencies, and includes five key milestones: Dataset Ready (end of Week 3), NLP Pipeline Complete (end of Week 7), Backend Complete (end of Week 9), Integrated Prototype Complete (end of Week 10), and Final Submission Ready (end of Week 12).

The NLP pipeline represents the critical path: backend development, frontend integration, and all subsequent phases are dependent on its completion. Parallel workstreams in the early weeks are used to compress the overall schedule where dependencies permit.

3.6 Project Resources

Human Resources

The project requires four team members contributing approximately 8–10 hours per week each across 12 weeks, giving a total estimated effort of 384–480 person-hours. All four members are expected to contribute to testing, documentation, and the final presentation regardless of primary role.

Technical Resources

A full breakdown of the tools and technologies used across each stage of the pipeline is provided in [Section 4.1](#).

Standard personal laptops will be used for all development work. Google Colab (free tier) will be available for NLP model training if local computing is insufficient; model selection will be constrained to run within available resources where possible. No specialised hardware is required.

4 Methodology

4.1 Required Tools and Technologies

Category	Tools	Purpose
Data collection	Reddit API, PRAW, Python	Collect Reddit posts and comments
Supplementary data	ABS 2021 Census datasets	Provide suburb background context if Reddit data is insufficient
API access	Reddit script app, client_id, client_secret, user_agent	Authenticate Reddit API access
Data cleaning	Pandas, Regex	Clean and prepare text data
Suburb detection	Target suburb list, full Melbourne suburb list, Regex	Match suburbs and reduce incorrect attribution
Topic detection	Keyword lists, Regex	Classify discussion topics
Sentiment analysis	VADER	Generate sentiment scores and labels
Data storage	PostgreSQL, SQL	Store raw and processed data
Backend	Flask, JSON	Serve processed results to frontend
Frontend	React, HTML5, CSS3, JavaScript	Build dashboard interface
Visualisation	Chart.js	Display charts and rankings
Version control	GitHub	Manage code collaboration
Containerisation	Docker	Provide consistent backend environment
Deployment	Vercel, Render	Host frontend and backend
Hardware/compute	Personal laptops, optional Google Colab	Support development and exploratory analysis

Table 6: List of Required Tools and Technologies

4.2 Data Source and Collection

The main data source is the public `r/melbourne` subreddit, collected through the Reddit API using PRAW in Python. PRAW is used because it provides Python-based access to Reddit API data (Boe, 2014). Reddit is selected because it contains public discussions about Melbourne suburbs, including housing affordability, safety, amenities, education, employment, and local living experiences.

For each selected suburb, the system will search `r/melbourne` using the suburb name as the keyword. A post will be kept only if the suburb appears in the post title or body. Collected fields will include **post ID, title, body, relevant comments, number of comments, created date, subreddit, and permalink**.

Two suburb lists will be used. The **target suburb list** contains the suburbs included in the dashboard, while the **full Melbourne suburb list** is used to detect comments that mention other suburbs outside the project scope. This reduces incorrect attribution.

If Reddit data is insufficient for some suburbs, ABS 2021 Census datasets may be used as supplementary background context, not as a replacement for Reddit sentiment analysis. ABS Census data can provide additional suburb-level information such as population, housing, income, education, and employment characteristics (Australian Bureau of Statistics, n.d.).

4.3 Data Pre-processing

Collected Reddit text will be cleaned before analysis. This includes lowercasing text, removing URLs and unnecessary symbols, handling missing or deleted content, and removing duplicates. The cleaned dataset will then be used for suburb matching, topic detection, and sentiment analysis.

4.4 Algorithms and Analysis Approach

This project uses rule-based text processing and lexicon-based sentiment analysis because the topic categories are predefined and the method needs to be transparent and repeatable.

Suburb Detection

```
FOR each matched post:
    main_suburb = suburb found in post title/body
    FOR each comment:
        IF comment mentions main_suburb:
            assign comment to main_suburb
        ELSE IF comment mentions another suburb in the target list:
            assign comment to that target suburb
        ELSE IF comment mentions another Melbourne suburb outside the target list:
            exclude comment from suburb-level analysis
        ELSE:
            assign comment to main_suburb
```

Topic Detection

Topic detection will use predefined keyword lists. Each record may receive one or more topic labels. If no keyword is matched, it will be labelled **General**.

Topic	Example Keywords
Lifestyle amenities	cafe, restaurant, park, shopping, supermarket, nightlife, quiet, convenient
Education and employment	school, university, campus, job, work, office, employment
Crime and safety	safe, unsafe, crime, theft, robbery, police, dangerous, security
Affordability	rent, expensive, cheap, price, cost, affordable, overpriced

Table 7: Example Keywords used for Topic Extraction

Sentiment Analysis

VADER will calculate a compound sentiment score for each cleaned record. VADER is selected because it is a lexicon and rule-based sentiment analysis method designed for social media text (Hutto & Gilbert, 2014). The compound score will be classified using standard VADER thresholds (VADER Sentiment, n.d.).

Compound Score	Sentiment Label
≥ 0.05	Positive
> -0.05 and < 0.05	Neutral
≤ -0.05	Negative

Table 8: Sentiment Score Threshold

The average compound score will be used as the overall suburb sentiment score. Key concerns will be calculated from topic frequencies in negative records, while key strengths will be calculated from topic frequencies in positive records.

4.5 System Architecture and Data Management

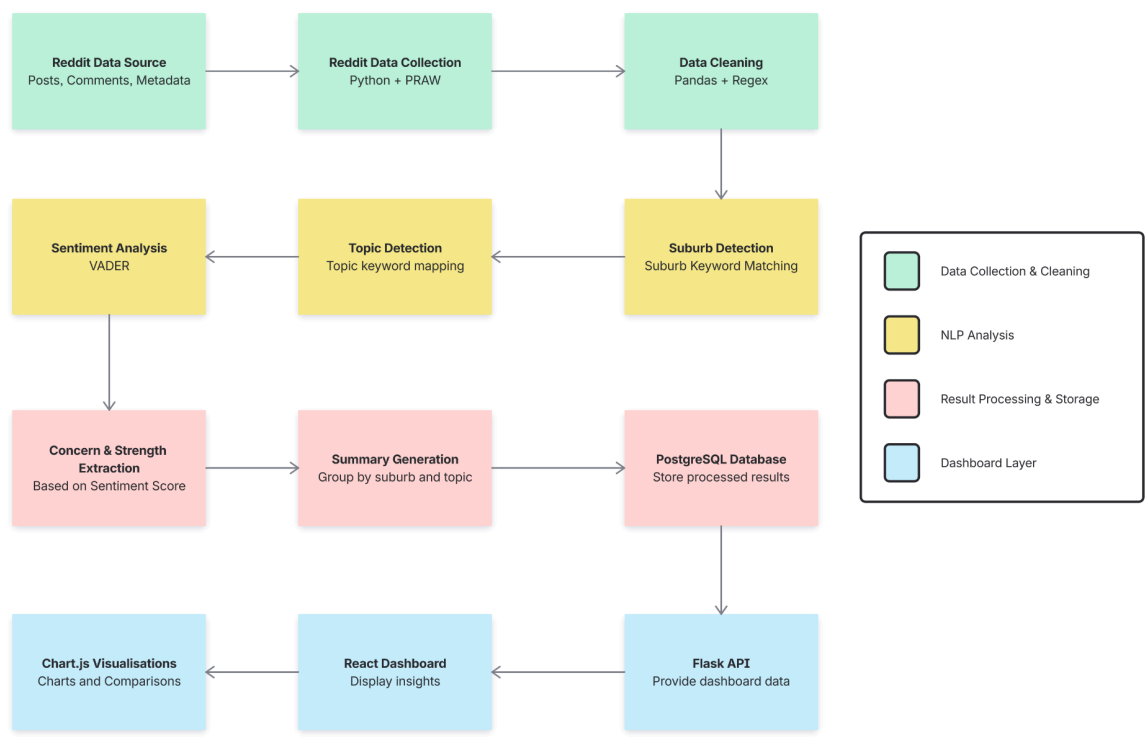


Figure 1: System Architecture

PostgreSQL will store raw and processed data in tables such as posts, comments, analysis_results, and suburb_summary. Flask will retrieve processed results from PostgreSQL and return JSON responses to the React frontend. Flask is suitable because it is a lightweight web framework for building web applications and APIs (Pallets, n.d.). Chart.js will display the dashboard visualisations. Docker will support a consistent backend environment, while Vercel and Render will be used for deployment. A dashboard wireframe is provided in [Appendix E](#) to show the planned user interface layout and key dashboard components.

This methodology is replicable because it defines the same data source, target suburb list, full suburb checking list, topic keywords, sentiment thresholds, database structure, software pipeline, and dashboard design direction.

5 Conclusion

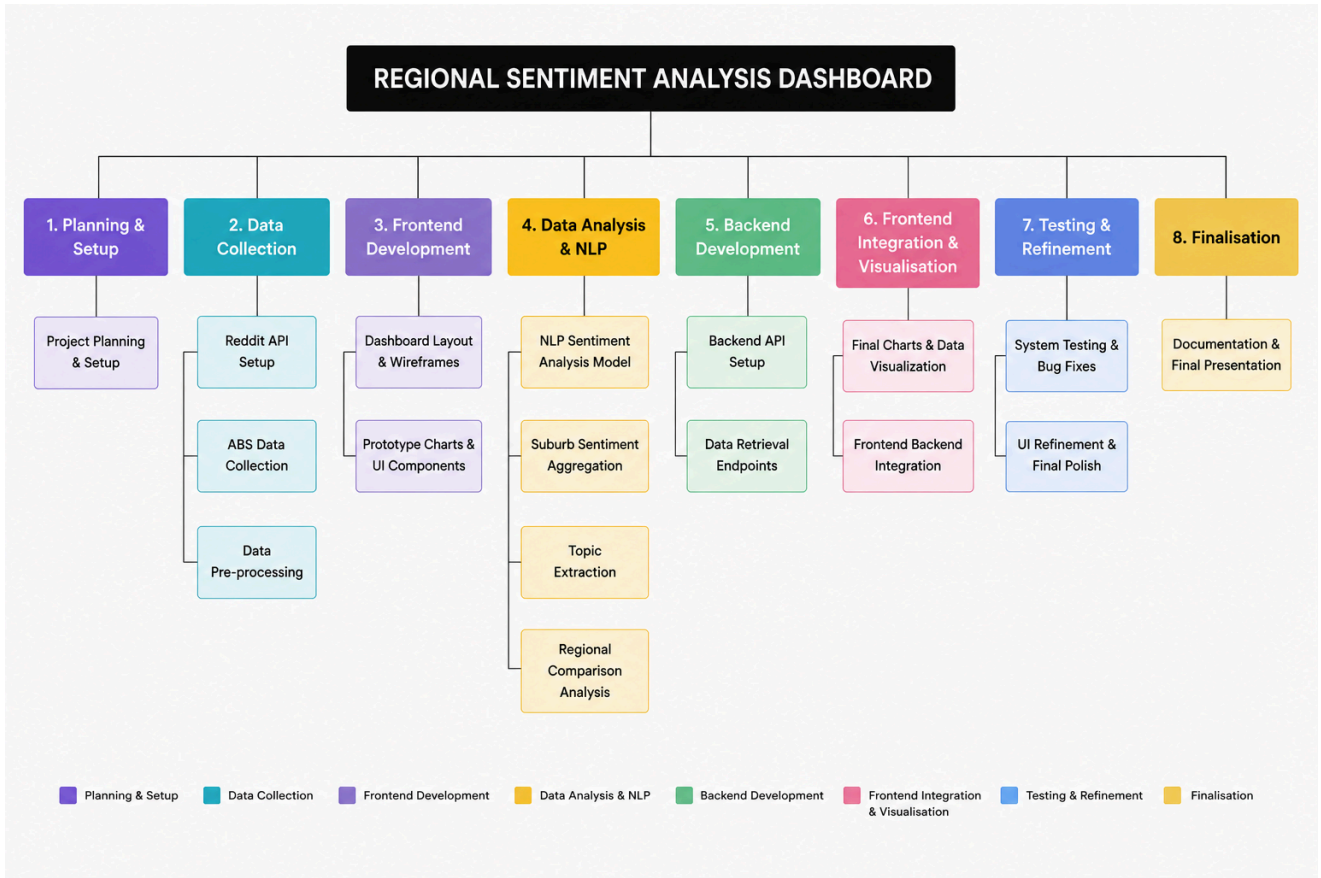
This report has proposed **Social Media Regional Housing Intelligence**, a data science project designed to support Melbourne-based homebuyers aged 30 to 45 when comparing selected Melbourne suburbs. The project responds to the limitation of traditional housing information, which often focuses on property prices, listings, and demographic indicators, but may not clearly reflect how people perceive everyday suburb experiences. By focusing on public Reddit discussions, the project aims to capture more local and perception-based insights about factors such as affordability, safety, lifestyle amenities, education, and employment access.

The proposed system addresses the research gap identified in the literature review, where existing studies explore geographic sentiment or the relationship between sentiment and housing prices, but provide limited detail on the specific local factors behind positive or negative suburb sentiment. To address this, the system will collect public posts and comments from `r/melbourne`, clean and preprocess the text, detect suburb mentions, classify discussions into topic categories, and apply VADER sentiment analysis. The analysed results will then be stored in PostgreSQL and displayed through a web-based dashboard using Flask, React, and Chart.js.

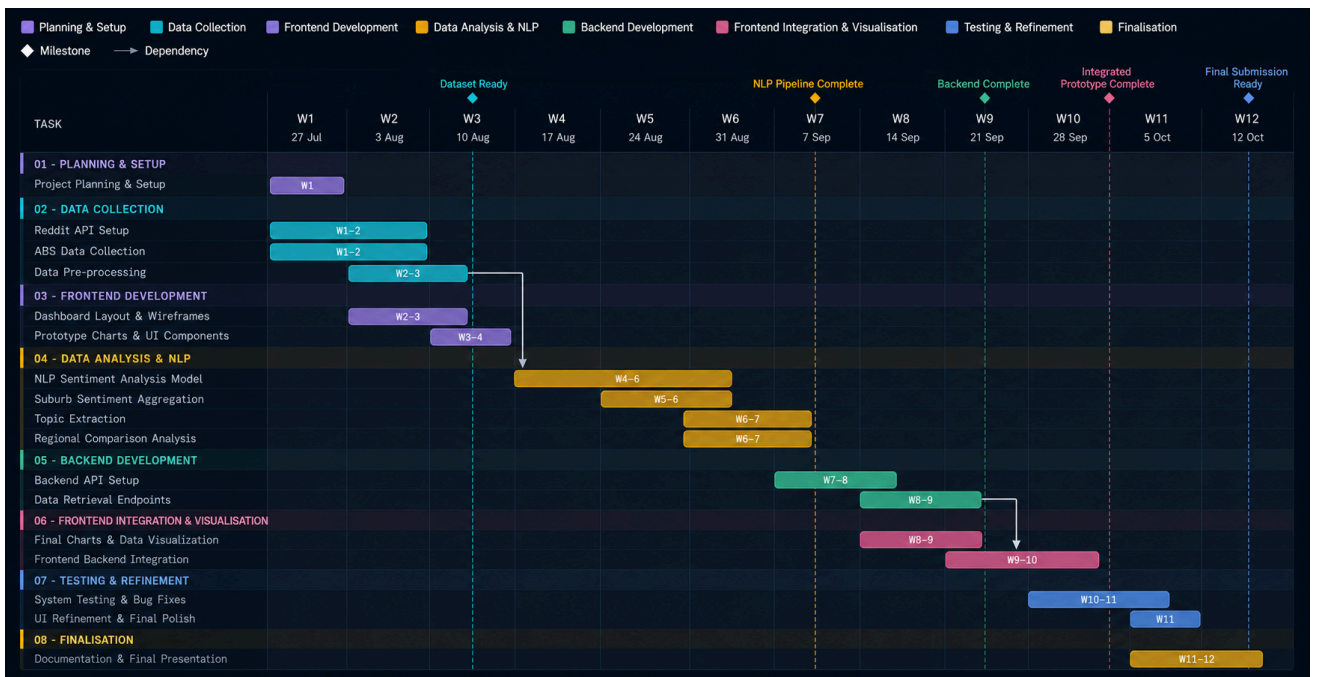
Overall, the expected outcome is an interactive dashboard that complements traditional property and demographic data with public sentiment insights. The dashboard will allow users to view sentiment scores, topic breakdowns, post and comment counts, key concerns, key strengths, and suburb comparisons in a clear visual format. While the project is not intended to replace professional property advice or official housing data, it can support more informed and context-aware decision-making by helping homebuyers better understand how selected Melbourne suburbs are discussed by the public.

6 Appendix

Appendix A: Work Breakdown Structure (WBS)



Appendix B: Gantt Chart



Appendix C: Requirements Traceability Matrix (RTM)

Req ID	Description	Categories	Type(s) (FR/NFR)	Source	Status
01	The software must be able to specify the number of posts or comments used for calculating sentiment score for each category in each suburb (eg. number of google reviews). This is done so that people can compare the legitimacy of each suburb sentiment score, naturally higher number of post/comments indicates higher legitimacy of the sentiment score	Calculating	FR	Survey	Not Started
02	The software must be able specify sentiment score for the topic category (eg. Lifestyle Amenities, Safety & Crime, Educational & Employment - tbc) in each suburb	Calculating	FR	Survey, Independent Research	Not Started
03	The software must be able to specify overall sentiment score for each suburb (Average of all of the sentiment scores for each topic category). This is done so that the user can summarise and compare the higher level view of the suburb	Calculating	FR	Survey	Not Started
04	Allow users to select and compare 2 suburbs simultaneously. This is done so that the user can conveniently decide.	Analysis	FR	Survey	Not Started
05	The system shall allow users to filter insights by topic		FR	Survey	Not Started
06	The system shall rank suburbs based on their sentiment score.		FR	Survey	Not Started
07	The software must be able to show sentiment comparison between suburbs in one page.	UI/UX	NFR	Survey	Not Started
08	The software must be able to show an overall sentiment score for each suburb when searched for.	UI/UX	NFR	Survey	Not Started
09	The software must be able to show the number of posts/sentiment for each suburb when searched for.	UI/UX	NFR	Survey	Not Started
10	The UI should be intuitive and easy to navigate for non-technical users (eg. The targeted demographic which is 30-45 y.o Australian)	UI/UX	NFR	Survey	Not Started

Appendix D: Risk Register

ID	Risk	Likelihood	Impact	Rating	Mitigation Strategy
R1	Insufficient Reddit data for some suburbs	Medium	High	High	Use nearby suburb grouping and supplementary ABS data
R2	Sentiment analysis accuracy below target (>80%)	Medium	High	High	Build baseline VADER model and iteratively improve performance
R3	Delays in dashboard development	Medium	Medium	Medium	Track progress through Trello and reassign tasks if required
R4	Team member unavailable due to university commitments	Medium	Medium	Medium	Maintain shared documentation and redistribute workload
R5	Integration issues between Flask backend and React frontend	Medium	High	High	Conduct early integration testing and frequent code reviews
R6	Data preprocessing errors affecting results	Medium	High	High	Perform validation and preprocessing testing regularly
R7	Schedule overruns near project deadline	Medium	High	High	Monitor milestones using Gantt Chart and prioritise critical features
R8	Loss of project files or code	Low	High	Medium	Use GitHub version control and regular backups
R9	Dashboard performance issues with large datasets	Medium	Medium	Medium	Conduct performance and load testing
R10	Requirement misunderstandings between team members	Low	Medium	Low	Regular meetings and documentation of decisions

Appendix E: Wireframe

Overall View

Social Media
Regional Housing
Intelligence

Overall View

Suburb View

Overall View

Topic Selection (Filter insights by topic)

All Topics

Lifestyle Amenities

Crime & Safety

Educational & Employment

Affordability

Suburb Ranking

Rank	Suburb	Sentiment Score	Sentiment	Action
1	Collingwood	+0.92	Positive	View Detail >
2	Oakleigh	+0.73	2	View Detail >
3	Nunawading	+0.52	3	View Detail >
4	Eaglemont	+0.34	4	View Detail >
5	Richmond	+0.12	5	View Detail >
6	Carlton	-0.15	6	View Detail >
7	Vermont	-0.34	7	View Detail >
8	Dandenong	-0.57	8	View Detail >
9	Officer	-0.86	9	View Detail >
10	Clayton	-0.88	10	View Detail >

Suburb View

Social Media
Regional Housing
Intelligence

Overall View

Suburb View

< Return

Year filter

Collingwood

+0.92/1.0

Positive (Number of posts: x)

Positive x%

Neutral x%

Negative x%

Top Strengths

Lifestyle Amenities (Number of posts: x)

(word 1, word 2, word 3)

Educational & Employment (Number of posts: x)

(word 1, word 2, word 3)

Top Concerns

Crime & Safety (Number of posts: x)

(word 1, word 2, word 3)

Oakleigh

+0.77/1.0

Positive (Number of posts: x)

Positive x%

Neutral x%

Negative x%

Top Strengths

Lifestyle Amenities (Number of posts: x)

(word 1, word 2, word 3)

Crime & Safety (Number of posts: x)

(word 1, word 2, word 3)

Top Concerns

Educational & Employment (Number of posts: x)

(word 1, word 2, word 3)

Suburb Dropdown List View

Social Media
Regional Housing
Intelligence

Overall View

Suburb View

< Return

Year filter

Collingwood ^

Oakleigh

Nunawading

Eaglemont

Richmond

Carlton

Vermont

Dandenong

Officer

Clayton

Lifestyle Amenities (Number of posts: x)
(word 1, word 2, word 3)

Educational & Employment (Number of posts: x)
(word 1, word 2, word 3)

Top Concerns

Crime & Safety (Number of posts: x)
(word 1, word 2, word 3)

Oakleigh v

+0.77/1.0

Positive (Number of posts: x)

Positive x%

Neutral x%

Negative x%

Top Strengths

Lifestyle Amenities (Number of posts: x)
(word 1, word 2, word 3)

Crime & Safety (Number of posts: x)
(word 1, word 2, word 3)

Top Concerns

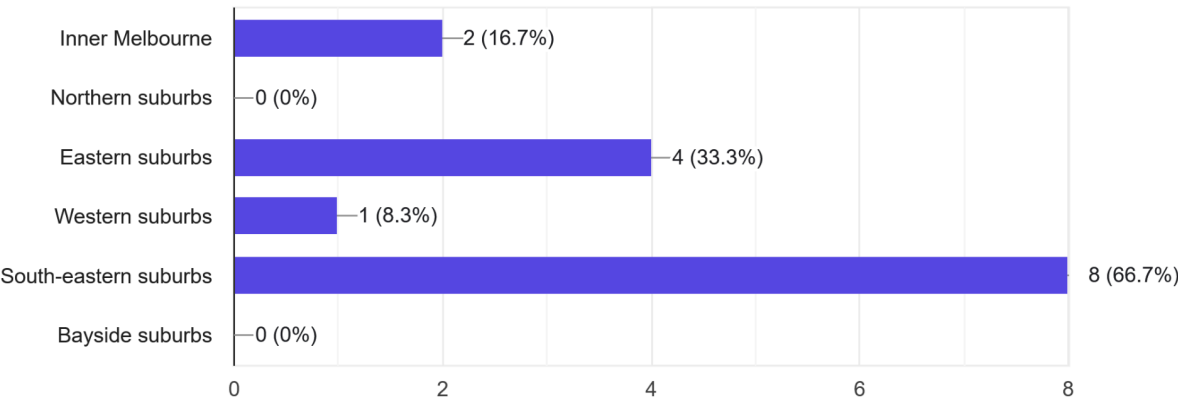
Educational & Employment (Number of posts: x)
(word 1, word 2, word 3)

Appendix F: User Survey

Suburb Selection

Which areas are you considering?

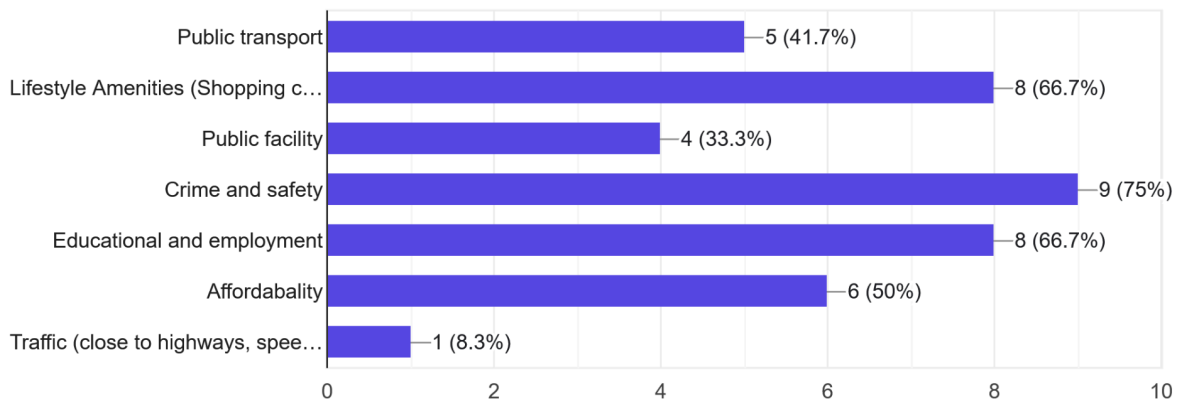
12 responses



Topic Selection

What are the main things you look for in a neighbourhood when you're thinking about buying a house?

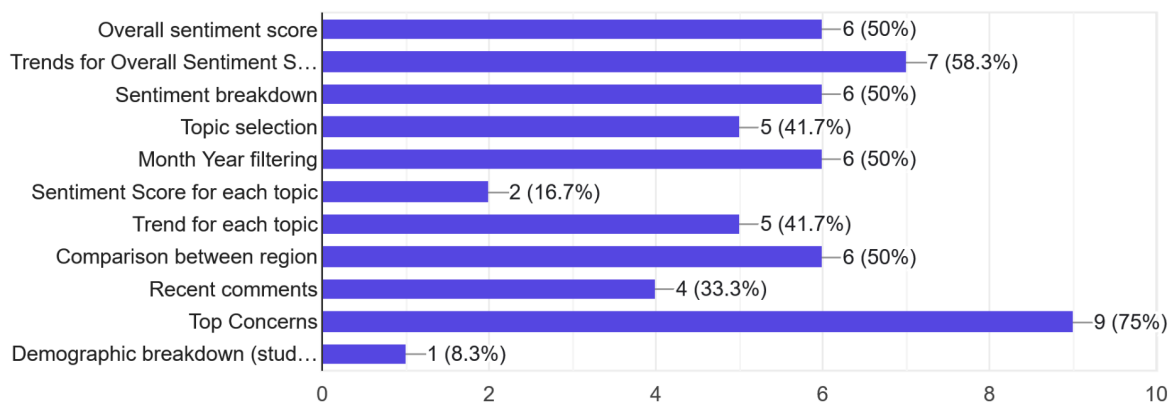
12 responses



Suburb View Feature Choice

What feature would you want to see in this tool for Region View? (Select all the apply)

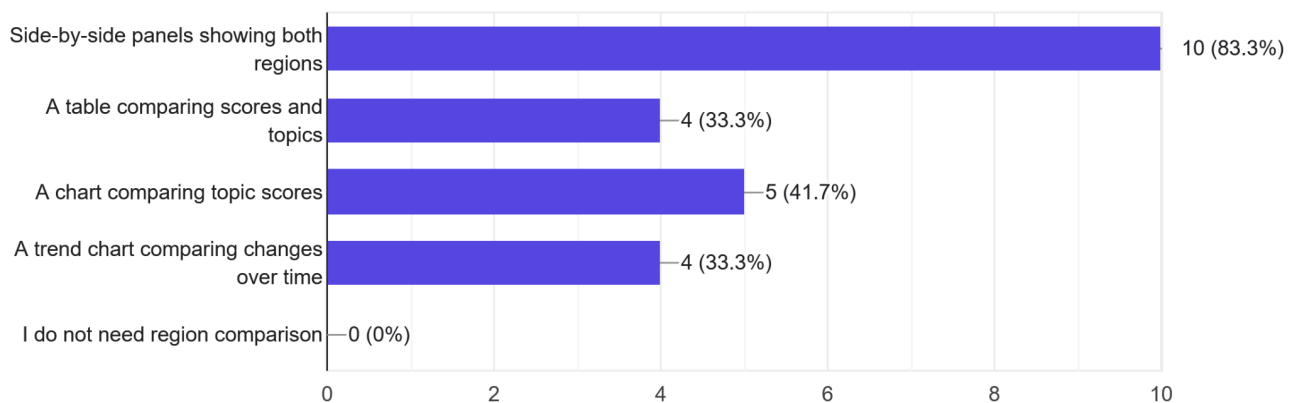
12 responses



Suburb Comparison View Choice

Would you like to compare two regions? If yes, how would you prefer?

12 responses



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